Ch – 3 Matrices

Concept, notation, order, equality, types of matrices, zero and identity matrix, transpose of a matrix, symmetric and skew symmetric matrices. Operations on matrices: Addition and multiplication and multiplication with a scalar. Simple properties of addition, multiplication and scalar multiplication. Non-commutativity of multiplication of matrices and existence of non-zero matrices whose product is the zero matrix (restrict to square matrices of order 2). Invertible matrices and proof of the uniqueness of inverse, if it exists; (Here all matrices will have real entries).

Matrix

A set of m×n numbers (weather real or complex) or functions arranged in the form of a rectangular array of m horizontal lines (called rows) and n vertical lines (called columns) is called a m × n matrix. A matrix is generally denoted by capital letter of English alphabet and an element of a matrix is denoted by a small letter. A matrix with m rows and n columns is called a matrix of order m by n.

Remark : (i) A matrix is an arrangement of numbers or functions and not a value.

(ii) Order of a matrix is written as m $\times n$ (read as m by n).

(ii) A general matrix of order m by n is written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

i = 1, 2, 3, ……….., m and j = 1, 2, 3, ……….., n

a_{ij} is called the (i, j)th element of the matrix A.

Types of matrices:

1. Row matrix: A matrix having only one row is called a row matrix.

e.g. A = $[1 \quad 3 \quad 5]$ is a row matrix of order 1× 3.

2. Column matrix: A matrix having only one column is called a column matrix. e.g. A = $\begin{bmatrix}1\\2\\3\end{bmatrix}$ is a column matrix of order 3 × 1.

3. Zero or Null matrix : A matrix is said to be a zero matrix if all its elements are zero.

i.e. A = $[a_{ij}]$ is null matrix if $a_{ij} = 0$ for all i,j.

Example: O = $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$, O = $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

4. Square matrix: A matrix A = $[a_{ij}]_{mxn}$ is said to be a square matrix if the no. of rows and columns in the matrix are same.

Example: A = $\begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix}$, A = $\begin{bmatrix} 2 & 0 & 5 \\ 0 & 5 & 6 \\ 4 & 0 & 8 \end{bmatrix}$

5. Diagonal elements of a square matrix : The elements a_{ij} are called diagonal elements of a square matrix A = $[a_{ij}]_{mxn}$ if i = j. i.e. a_{11}, a_{22}, a_{33}, ...,a_{nn} are called the diagonal elements of a square matrix. The line containing the elements a_{11}, a_{22}, a_{33}, ...,a_{nn} is called the principal diagonal.

6. Diagonal matrix: A square matrix is said to be a diagonal matrix if its non- diagonal elements are zeros.

e.g. A = $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 8 \end{bmatrix}$ is a diagonal matrix.

7. Scalar Matrix:

A square matrix is said to be a scalar matrix if its non- diagonal elements are zeros and all diagonal elements are equal.

As for example : A = $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

8.Unit matrix : A diagonal matrix is called a unit matrix if all the diagonal elements are unity.

As for example : $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

9.Equal Matrices: Two matrices A = $[a_{ij}]_{mx\,n}$ and B = $[b_{ij}]_{mx\,n}$ of the same order are equal if $a_{ij} = b_{ij}$ for all i, j

10. Upper Triangular matrix: A square matrix A = $[a_{ij}]_{nxn}$ is called an upper triangular matrix iff a_{ij}= 0 for all i > j i.e. all elements below the principal diagonal are zero.

As for example : A = $\begin{bmatrix} 2 & 3 & 6 \\ 0 & 2 & 7 \\ 0 & 0 & 2 \end{bmatrix}$

11.Lower Triangular matrix: A square matrix A = $[a_{ij}]_{nxn}$ is called an lower triangular matrix iff a_{ij}= 0 for all i < j i.e. all elements above the principal diagonal are zero.

As for example : A = $\begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 0 \\ 1 & 3 & 2 \end{bmatrix}$

12.Triangular matrix: A square matrix A = $[a_{ij}]_{nxn}$ is called a triangular matrix if it is either upper triangular or lower triangular.

Addition of matrices

Let A and B be two matrices of the same order m $\times n$. Then their sum denoted by A + B is also a matrix of the same order m $\times n$ and is obtained by adding the corresponding elements of A and B.

If A = $[a_{ij}]_{m \times n}$ and B = $[b_{ij}]_{m \times n}$ then A + B = $[a_{ij} + b_{ij}]_{m \times n}$

As for example :

If A = $\begin{bmatrix} 8 & 7 \\ 6 & 2 \end{bmatrix}$, B = $\begin{bmatrix} 5 & 1 \\ 9 & 0 \end{bmatrix}$, then A + B = $\begin{bmatrix} 13 & 8 \\ 15 & 2 \end{bmatrix}$,

Properties of matrix addition

If A = $[a_{ij}]_{m \times n}$ and B = $[b_{ij}]_{m \times n}$ are of the same order then

(i) Matrix addition is commutative i.e. A + B = B + A
(ii) Matrix addition is associative i.e. (A + B) + C = A + (B + C)
(iii) Existence of additive Identity : If A is any matrix of order m $\times n$ then there exists a null matrix O of order m $\times n$ such that A + O = O + A = A
(iv) Existence of additive Inverse : A + (-A) = (-A) + A = O
(v) Cancellation Laws: A + B = A + C ⇒B =C and B + A = C + A ⇒ B =C

Properties of scalar multiplication

If A and B are two matrices of the same order and k, l are scalars then

(i) k(A + B) = kA + k B
(ii) (-k)A = -(k A) = k (-A)
(iii) I A = A
(iv) (-1) A = - A

Difference of two matrices

Let A = $[a_{ij}]_{m \times n}$ and B = $[b_{ij}]_{m \times n}$ be two matrices of the same order m $\times n$. Then their difference denoted by A $-$ B is also a matrix of the same order m $\times n$ and is defined as

$$A - B = [a_{ij} - b_{ij}]_{m \times n}$$

Multiplication of Matrices

Two matrices A and B are said to be conformal for the product if the no. of columns in A is equal to the no. of rows in B.

Let A = $[a_{ij}]_{m \times n}$ and B = $[b_{jk}]_{m \times n}$ be two matrices then the product of A and B is a matrix C = $[c_{ik}]_{m \times p}$ where

$$c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \ldots\ldots + a_{in}b_{nk} = \sum_{j=1}^{n} a_{ij}b_{jk}$$

Properties of matrix multiplication

(i) Matrix multiplication is not commutative i.e. AB ≠ BA
(ii) Matrix multiplication is associative i.e. (AB)C = A(BC)
(iii) Matrix multiplication is distributive over matrix addition i.e. A(B+C) = AB + AC

(iv) If A is an $m \times n$ matrix and I is the identity matrix of order $n \times n$ then I A = A = A I
(v) If A is an $m \times n$ matrix and O is the null matrix of order $n \times p$ then A O = O
(vi) In a matrix multiplication the product of two non-zero matrices may be a zero matrix.

Transpose of a matrix The transpose of a matrix A is denoted by A^T or A' which is obtained by interchanging its rows and columns.

Properties of Transpose of a Matrix

(i) $(A^T)^T = A$ (iii) $(A+B)^T = A^T + B^T$
(ii) $(KA)^T = K A^T$ (iv) $(AB)^T = B^T A^T$
(v) $(ABC)^T = C^T B^T A^T$

Symmetric matrix A square matrix A = $[a_{ij}]_{n\times n}$ is said to be a symmetric matrix if transpose of A is equal to matrix A. i.e. if $A^T = A$.

For example: Let A= $\begin{bmatrix} 2 & 5 \\ 5 & 3 \end{bmatrix}$, then $\begin{bmatrix} 2 & 5 \\ 5 & 3 \end{bmatrix}^T = \begin{bmatrix} 2 & 5 \\ 5 & 3 \end{bmatrix}$

∴ A is said is symmetric matrix.

Skew-symmetric matrix A square matrix A = $[a_{ij}]_{n\times n}$ is said to be a skew symmetric matrix if $A^T = -A$.

For example:

Let A = $\begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$, then $A^T = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} = - \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix} = -A$

∴ A is skew-symmetric matrix.

Note:

(1) All main diagonal elements of a skew symmetric matrix are zero.
(2) Every square matrix can be uniquely expressed as a sum of a symmetric and a skew-symmetric matrix.
(3) All positive integral powers of a symmetric matrix are symmetric.
(4) odd positive integral powers of a skew-symmetric matrix are skew-symmetric.

Invertible matrices

Let A be a square matrix of order n, If then there exist a square matrix B such that $AB = I_n = BA$, then B is called the inverse of A and is denoted by A^{-1}.

$$\therefore AA^{-1} = I = A^{-1}A$$

If A and B are invertible square matrices of the same order then AB is also invertible and $(AB)^{-1} = B^{-1}A^{-1}$